1 Convert $\frac{2}{9}$ to a decimal.

$$\begin{array}{r} 0.2222 \\ 9\overline{)2.0000} \end{array}$$

$0.\dot{2}$

(Total for question 1 is 2 marks)

2 Convert $\frac{4}{11}$ to a decimal.

$$\begin{array}{r} 0.36363 \\ 11\overline{)4.0000} \end{array}$$

$0.\dot{3}\dot{6}$

(Total for question 2 is 2 marks)

3 Convert $\frac{5}{6}$ to a decimal.

$$\begin{array}{r} 0.83333 \\ 6\overline{)5.0000} \end{array}$$

$0.8\dot{3}$

(Total for question 3 is 2 marks)

4 Prove algebraically that the recurring decimal $0.\dot{8}$ can be written as $\frac{8}{9}$

(1) $0.\dot{8} = x$

(2) $8.\dot{8} = 10x$

(2) - (1) $8 = 9x$

$x = 8/9$

(Total for question 4 is 2 marks)

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5 Prove algebraically that the recurring decimal $0.4\dot{7}$ can be written as $\frac{43}{90}$

$0.4\dot{7} = x$

(1) $4.\dot{7} = 10x$

(2) $47.\dot{7} = 100x$

(2) - (1) $43 = 90x$

$x = \frac{43}{90}$

(Total for question 5 is 2 marks)

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6 Prove algebraically that the recurring decimal $0.2\dot{3}$ can be written as $\frac{7}{30}$

$0.2\dot{3} = x$

(1) $2.\dot{3} = 10x$

(2) $23.\dot{3} = 100x$

(2) - (1) $21 = 90x$

$x = \frac{21}{90} = \frac{7}{30}$

(Total for question 6 is 2 marks)

7 Write $0.1\dot{6}$ as a fraction in its simplest form.

$0.1\dot{6} = x$

$1.\dot{6} = 10x$

$16.\dot{6} = 100x$

$15 = 90x$

$x = \frac{15}{90} = \frac{1}{6}$

$\frac{1}{6}$

(Total for question 7 is 2 marks)

8 Write $0.2\dot{7}$ as a fraction in its simplest form.

$0.2\dot{7} = x$

$2.\dot{7} = 10x$

$27.\dot{7} = 100x$

$25 = 90x$

$x = \frac{25}{90} = \frac{5}{18}$

$\frac{5}{18}$

(Total for question 8 is 2 marks)

9 Write $0.4\dot{3}$ as a fraction in its simplest form.

$0.4\dot{3} = x$

$4.\dot{3} = 10x$

$43.\dot{3} = 100x$

$39 = 90x$

$x = \frac{39}{90} = \frac{13}{30}$

$\frac{13}{30}$

(Total for question 9 is 2 marks)

10 Prove algebraically that the recurring decimal $0.6\dot{8}\dot{1}$ can be written as $\frac{15}{22}$

$$0.6\dot{8}\dot{1} = x$$
$$6.\dot{8}\dot{1} = 10x$$
$$681.\dot{8}\dot{1} = 1000x$$
$$675 = 990x$$
$$x = \frac{675}{990} = \frac{15}{22}$$

(Total for question 10 is 2 marks)

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11 Prove algebraically that the recurring decimal $0.\dot{2}1\dot{6}$ can be written as $\frac{8}{37}$

$$0.\dot{2}1\dot{6} = x$$
$$216.\dot{2}1\dot{6} = 1000x$$
$$216 = 999x$$
$$x = \frac{216}{999} = \frac{8}{37}$$

(Total for question 11 is 2 marks)

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12 Prove algebraically that the recurring decimal $0.\dot{1}2\dot{6}$ can be written as $\frac{14}{111}$

$$0.\dot{1}2\dot{6} = x$$
$$126.\dot{1}2\dot{6} = 1000x$$
$$126 = 999x$$
$$x = \frac{126}{999} = \frac{14}{111}$$

(Total for question 12 is 2 marks)

13 Write $3.2\dot{5}\dot{4}$ as a fraction in its simplest form.

$$3.2\dot{5}\dot{4} = x$$
$$32.\dot{5}\dot{4} = 10x$$
$$3254.\dot{5}\dot{4} = 1000x$$
$$3222 = 990x$$
$$x = \frac{3222}{990}$$
$$= \frac{179}{55} \text{ OR } 3\frac{14}{55}$$

$\frac{179}{55}$

(Total for question 13 is 3 marks)

14 Write $2.7\dot{4}\dot{2}$ as a fraction in its simplest form.

$$2.7\dot{4}\dot{2} = x$$
$$27.\dot{4}\dot{2} = 10x$$
$$2742.\dot{4}\dot{2} = 1000x$$
$$2715 = 990x$$
$$x = \frac{2715}{990}$$
$$= \frac{181}{66} \text{ OR } 2\frac{49}{66}$$

$\frac{181}{66}$

(Total for question 14 is 3 marks)

15 Write $3.\dot{5}9\dot{4}$ as a fraction in its simplest form.

$$3.\dot{5}9\dot{4} = x$$
$$3594.\dot{5}9\dot{4} = 1000x$$
$$3591 = 999x$$
$$x = \frac{3591}{999}$$
$$= \frac{133}{37} \text{ OR } 3\frac{22}{37}$$

$\frac{133}{37}$

(Total for question 15 is 3 marks)

16 x is an integer such that $1 \leq x \leq 9$

Prove that $0.\dot{0}\dot{x} = \frac{x}{99}$

① $0.\dot{0}\dot{x} = y$

② $\cancel{0}x.\dot{0}\dot{x} = 100y$

② − ① $x = 99y$

$y = \frac{x}{99}$

.....

(Total for question 16 is 2 marks)

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17 Work out: $0.\dot{5}\dot{4} \times 0.\dot{5}$

$0.\dot{5}\dot{4} = x$

$54.\dot{5}\dot{4} = 100x$

$54 = 99x$

$x = \frac{54}{99} = \frac{6}{11}$

$0.\dot{5} = y$

$5.\dot{5} = 10y$

$5 = 9y$

$y = \frac{5}{9}$

$\frac{6}{11} \times \frac{5}{9} = \frac{30}{99}$

$= \frac{10}{33}$

$\frac{10}{33}$

.....

(Total for question 17 is 4 marks)

18 Work out: $0.\dot{3}\dot{9} \div 0.\dot{6}\dot{3}$

$0.\dot{3}\dot{9} = x$

$39.\dot{3}\dot{9} = 100x$

$39 = 99x$

$x = \frac{39}{99}$

$= \frac{13}{33}$

$0.\dot{6}\dot{3} = y$

$63.\dot{6}\dot{3} = 100y$

$63 = 99y$

$y = \frac{63}{99}$

$= \frac{7}{11}$

$\frac{13}{33} \div \frac{7}{11}$

$\frac{13}{\cancel{33}_{3}} \times \frac{\cancel{11}^{1}}{7} = \frac{13}{21}$

$\frac{13}{21}$

(Total for question 18 is 4 marks)

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19 Work out: $0.0\dot{7} \div 0.\dot{1}8\dot{5}$

$0.0\dot{7} = x$

$0.\dot{7} = 10x$

$7.\dot{7} = 100x$

$7 = 90x$

$x = \frac{7}{90}$

$0.\dot{1}8\dot{5} = y$

$185.\dot{1}8\dot{5} = 1000y$

$185 = 999y$

$y = \frac{185}{999}$

$= \frac{5}{27}$

$\frac{7}{90} \div \frac{5}{27}$

$\frac{7}{\cancel{90}_{10}} \times \frac{\cancel{27}^{3}}{5} = \frac{21}{50}$

$\frac{21}{50}$

(Total for question 19 is 4 marks)